◆ CUSTOMER SERVICE REQUEST ANALYSIS

◆ 1. Import Libraries

Load required Python libraries like Pandas, NumPy, Matplotlib, Seaborn.

◆ 2. Load Dataset

Import dataset using CSV/Excel and preview data.

◆ 3. Understand Dataset

Check structure, data types, and summary statistics.

◆ 4. Data Cleaning

Handle missing values, duplicates, and incorrect formats.

◆ 5. Exploratory Data Analysis (EDA)

Analyze trends using charts and summaries.

◆ 6. Complaint Analysis

Identify common issues and service request patterns.

◆ 7. Statistical Testing

Perform hypothesis testing (p-value, Kruskal-Wallis).

◆ 8. Observations & Insights

Summarize key findings and business insights.

Import Libraries

In [22]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Read Dataset

In [2]:
```
import pandas as pd

df = pd.read_csv(
    "311_Service_Requests_from_2010_to_Present.csv",
    low_memory=False
)

df.isnull().sum()
```

Out[2]:
```
Unique Key                          0
Created Date                        0
Closed Date                      2381
Agency                              0
Agency Name                         0
Complaint Type                      0
Descriptor                       6501
Location Type                     133
Incident Zip                     2998
Incident Address                51699
Street Name                     51699
Cross Street 1                  57188
Cross Street 2                  57805
Intersection Street 1          313438
Intersection Street 2          314046
Address Type                     3252
City                             2997
Landmark                       364183
Facility Type                    2389
Status                              0
Due Date                            3
Resolution Description              0
Resolution Action Updated Date   2402
Community Board                     0
Borough                             0
X Coordinate (State Plane)       4030
Y Coordinate (State Plane)       4030
Park Facility Name                  0
Park Borough                        0
School Name                         0
School Number                       0
School Region                       1
School Code                         1
School Phone Number                 0
School Address                      0
School City                         0
School State                        0
School Zip                          1
School Not Found                    0
School or Citywide Complaint   364558
Vehicle Type                   364558
Taxi Company Borough           364558
Taxi Pick Up Location          364558
Bridge Highway Name            364261
Bridge Highway Direction       364261
Road Ramp                      364296
Bridge Highway Segment         364296
Garage Lot Name                364558
Ferry Direction                364557
Ferry Terminal Name            364556
Latitude                         4030
Longitude                        4030
Location                         4030
dtype: int64
```

In [3]:
```
df.head()
```

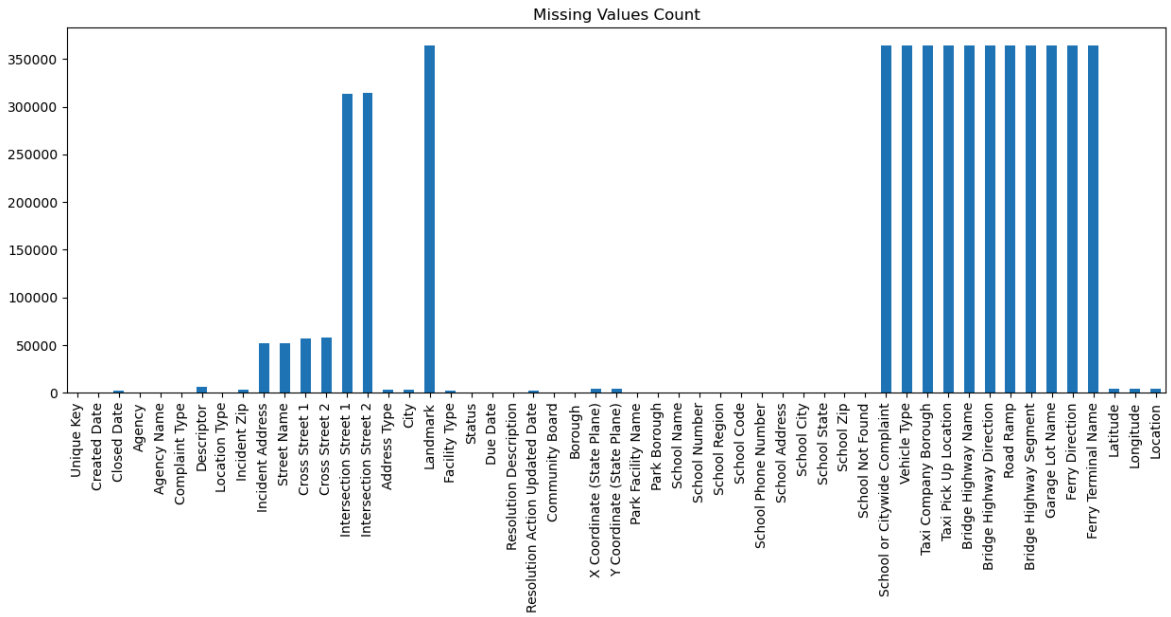
Out[3]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk

5 rows × 53 columns

Data Cleaning

In [4]:
```
df.isnull().sum().plot(kind='bar', figsize=(15,5))
plt.title("Missing Values Count")
plt.show()
```



In [5]:
```
df = df[df['Closed Date'].notnull()]
```

Convert date columns

In [6]:
```
df['Created Date'] = pd.to_datetime(df['Created Date'], format='%m/%d/%Y %I:%M:%
df['Closed Date'] = pd.to_datetime(df['Closed Date'], format='%m/%d/%Y %I:%M:%S
```

Remove incorrect timeline

In [7]:
```
df = df[df['Closed Date'] >= df['Created Date']]
```

Calculate Request Closing Time

In [8]:
```
df['Request_Closing_Time'] = df['Closed Date'] - df['Created Date']
```

Convert to seconds

In [9]:
```
df['Request_Closing_Time'] = (
    pd.to_datetime(df['Closed Date'], errors='coerce') -
    pd.to_datetime(df['Created Date'], errors='coerce')
).dt.total_seconds()
```

Descriptive stats

In [10]:
```
df['Request_Closing_Time'].describe()
```

Out[10]:
```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

Null check (Complaint Type & City)

In [11]:
```
df[['Complaint Type','City']].isnull().sum()
```

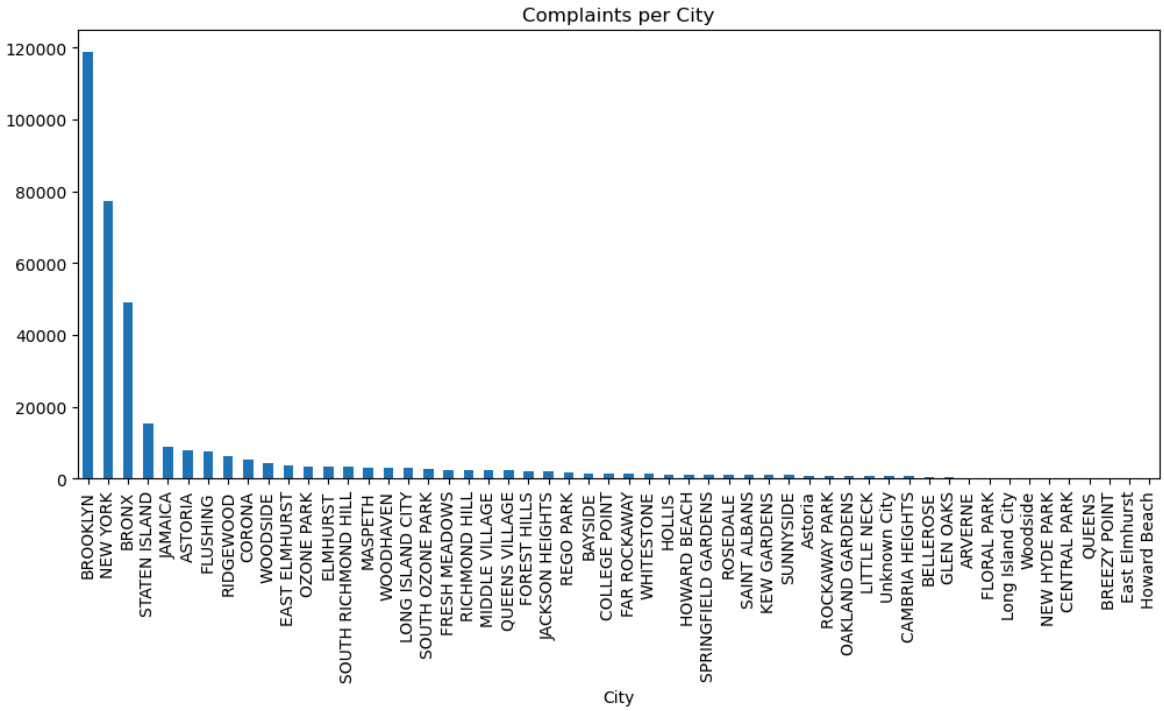
Out[11]:
```
Complaint Type      0
City              674
dtype: int64
```

Fill missing City

In [12]:
```
df['City'] = df['City'].fillna('Unknown City')
```

Frequency plot (complaints per city)

In [13]:
```
df['City'].value_counts().plot(kind='bar', figsize=(12,5))
plt.title("Complaints per City")
plt.show()
```

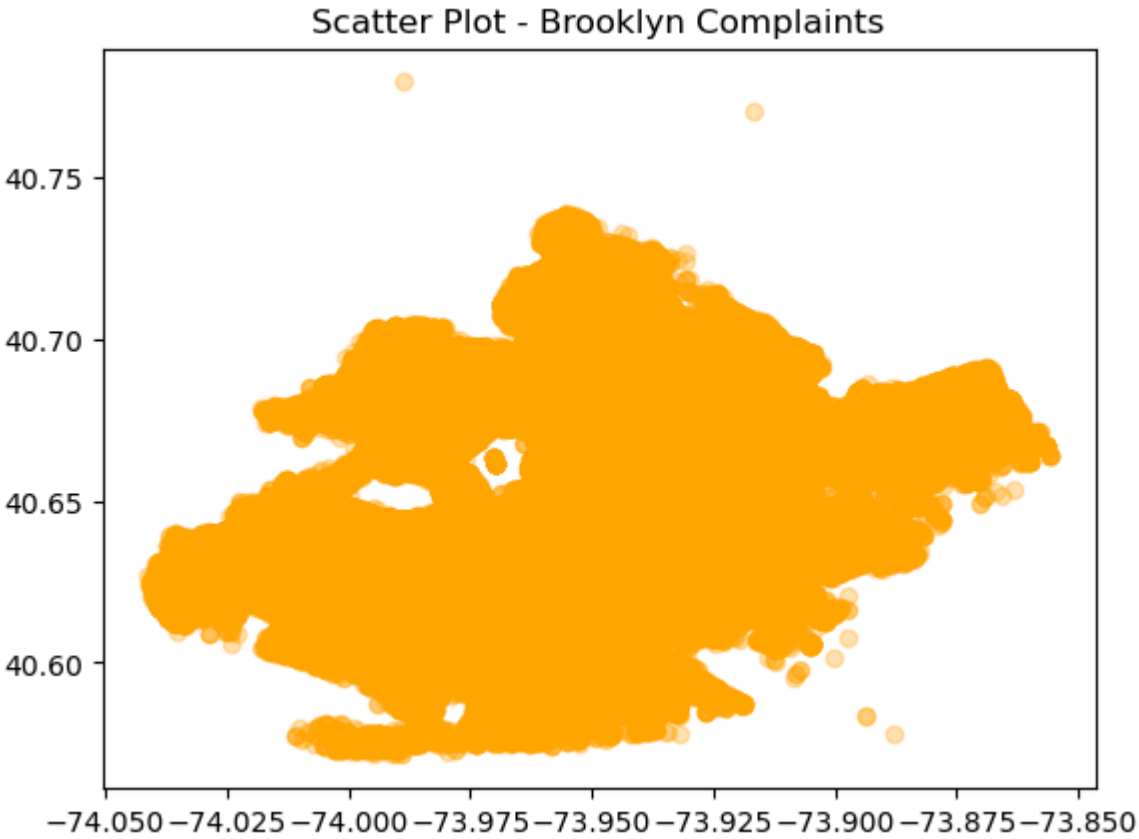


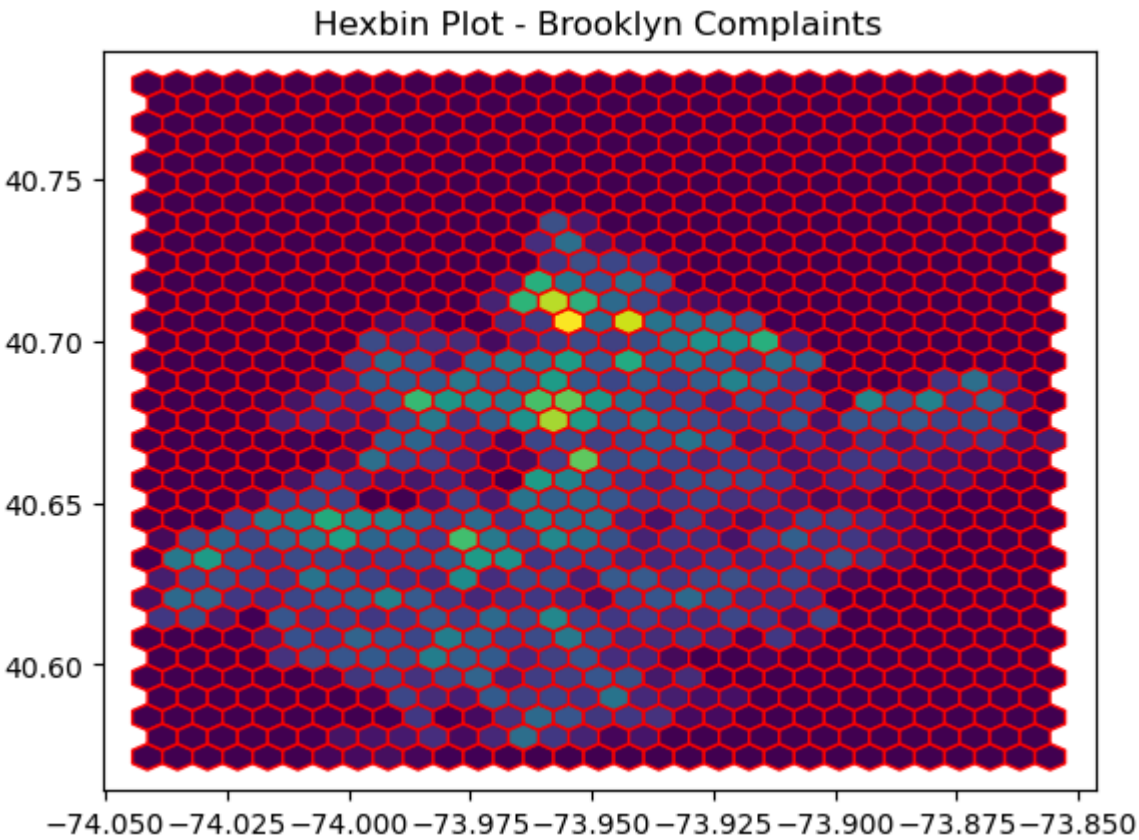
Scatter & Hexbin (Brooklyn)

In [14]:
```
brooklyn = df[df['City'] == 'BROOKLYN']

plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'], alpha=0.3,color ='orang
plt.title("Scatter Plot - Brooklyn Complaints")
plt.show()

plt.hexbin(brooklyn['Longitude'], brooklyn['Latitude'], gridsize=30,color ='red'
plt.title("Hexbin Plot - Brooklyn Complaints")
plt.show()
```

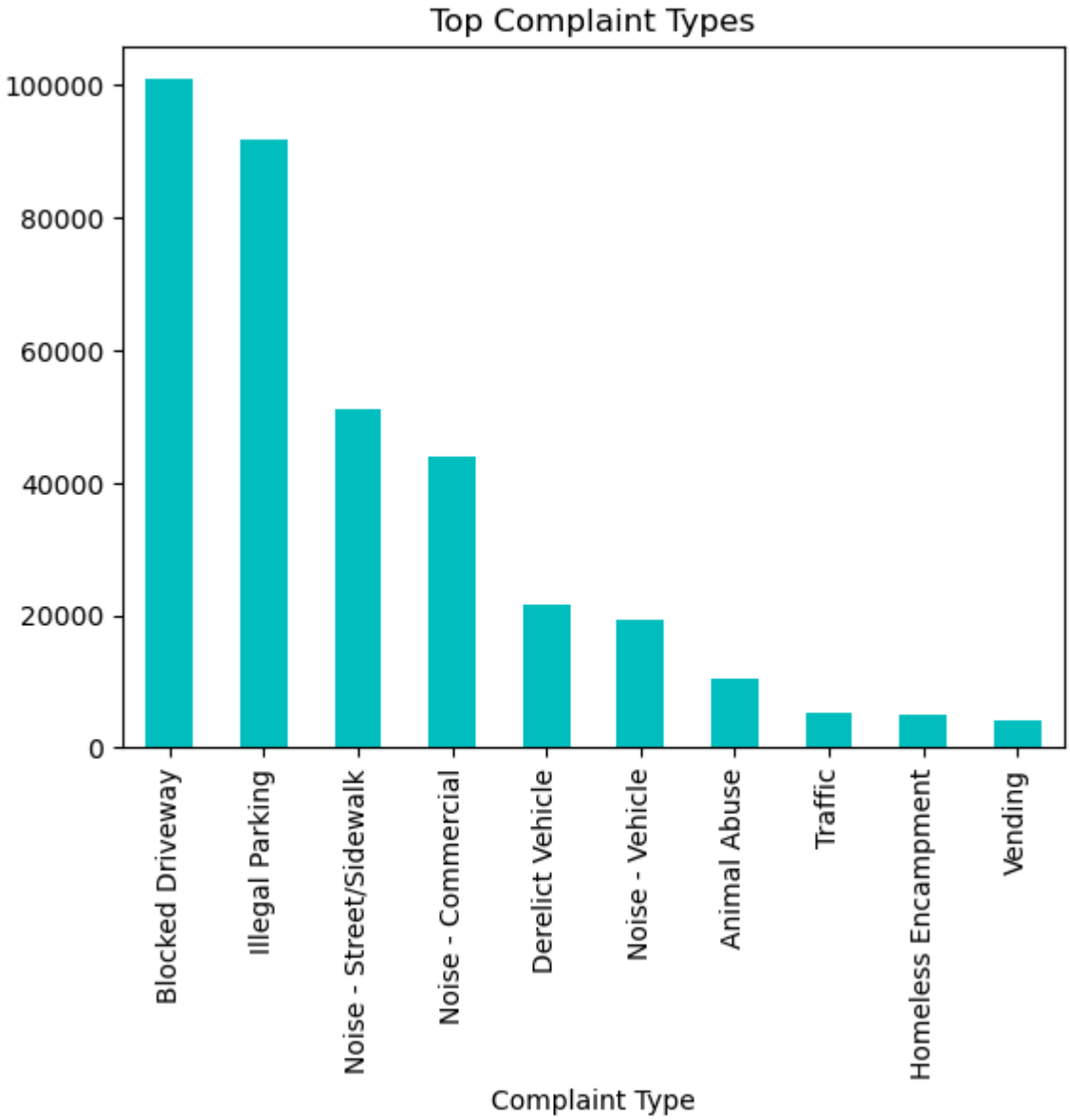




Major Complaint Types

Bar graph (complaint types

In [15]:
```
df['Complaint Type'].value_counts().head(10).plot(kind='bar',color='c')
plt.title("Top Complaint Types")
plt.show()
```



Frequency in NYC

In [16]:
```
nyc = df[df['City'] == 'NEW YORK']
nyc['Complaint Type'].value_counts().head(10)
```

Out[16]:
```
Complaint Type
Noise - Street/Sidewalk    22245
Noise - Commercial         18686
Illegal Parking            14549
Noise - Vehicle             6294
Homeless Encampment         3060
Blocked Driveway            2705
Vending                     2638
Animal Abuse                1941
Traffic                     1769
Noise - Park                1243
Name: count, dtype: int64
```

Top 10 complaints

In [17]:
```
top10 = df['Complaint Type'].value_counts().head(10)
print(top10)
```

```
Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```

Complaints per city

In [18]:
```
df.groupby('City')['Complaint Type'].value_counts()
```

Out[18]:
```
City      Complaint Type
ARVERNE   Illegal Parking             62
          Blocked Driveway            50
          Animal Abuse                46
          Derelict Vehicle            32
          Noise - Street/Sidewalk     29
                                     ...
Woodside  Illegal Parking            124
          Blocked Driveway            27
          Derelict Vehicle             8
          Noise - Street/Sidewalk      5
          Noise - Commercial           2
Name: count, Length: 792, dtype: int64
```

Create df_new (pivot table)

In [19]:
```
df_new = pd.pivot_table(df,
                        index='Complaint Type',
                        columns='City',
                        aggfunc='size',
                        fill_value=0)

df_new.head()
```

Out[19]:

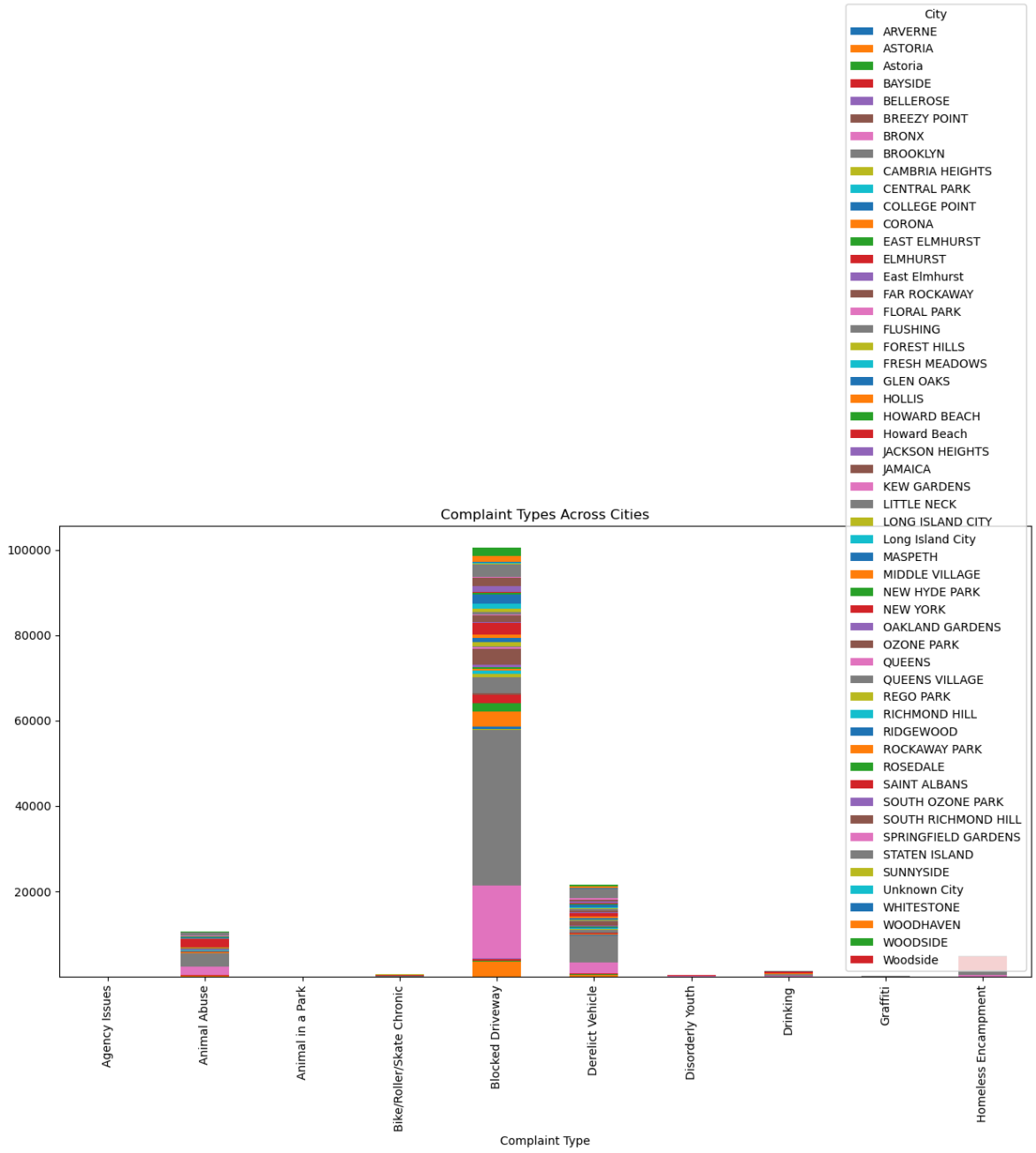
City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX
Complaint Type							
Agency Issues	0	0	0	0	0	0	0
Animal Abuse	46	170	0	53	15	2	1971
Animal in a Park	0	0	0	0	0	0	0
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22
Blocked Driveway	50	3436	159	514	138	3	17062

5 rows × 54 columns

Visualization

Complaints in each city

In [20]:
```
df_new.head(10).plot(kind='bar', stacked=True, figsize=(15,7))
plt.title("Complaint Types Across Cities")
plt.show()
```



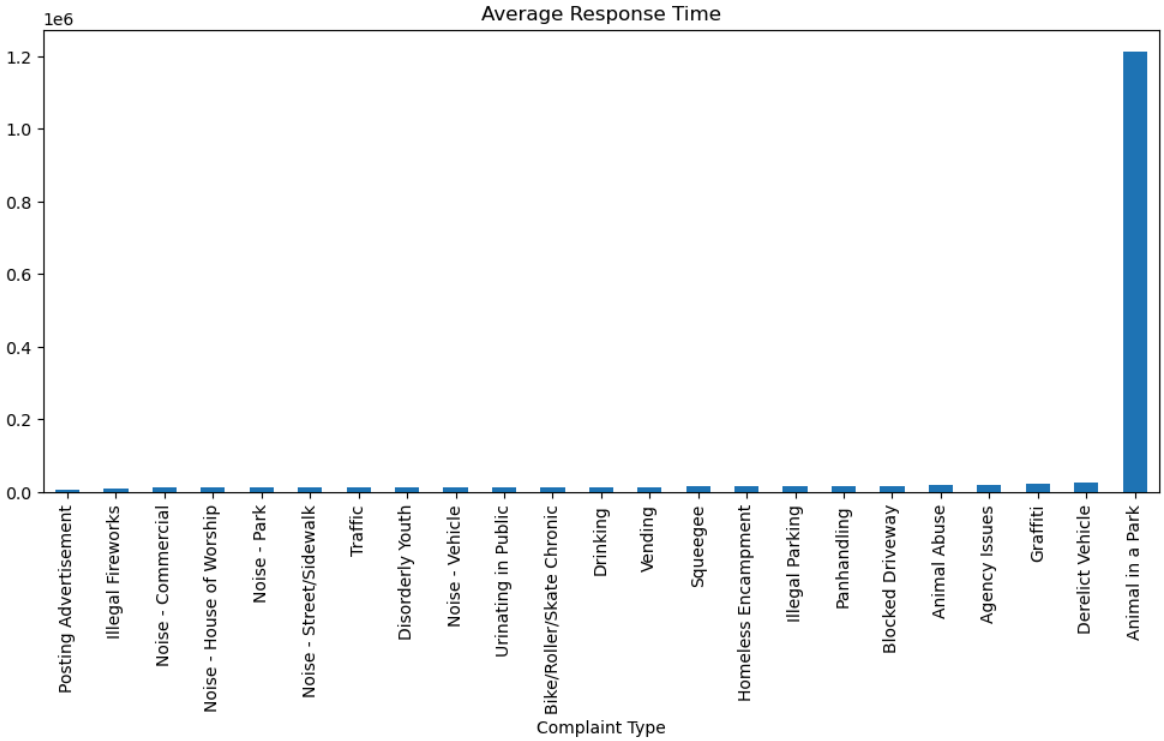
Sort by avg closing time

In [21]:
```
avg_time = df.groupby('Complaint Type')['Request_Closing_Time'].mean().sort_valu
print(avg_time)
```

```
Complaint Type
Posting Advertisement        7.286256e+03
Illegal Fireworks            1.011348e+04
Noise - Commercial           1.108576e+04
Noise - House of Worship     1.139109e+04
Noise - Park                 1.222606e+04
Noise - Street/Sidewalk      1.223130e+04
Traffic                      1.230912e+04
Disorderly Youth             1.236375e+04
Noise - Vehicle              1.256180e+04
Urinating in Public          1.295929e+04
Bike/Roller/Skate Chronic    1.312369e+04
Drinking                     1.382130e+04
Vending                      1.436628e+04
Squeegee                     1.456025e+04
Homeless Encampment          1.545138e+04
Illegal Parking              1.565044e+04
Panhandling                  1.585355e+04
Blocked Driveway             1.623252e+04
Animal Abuse                 1.803256e+04
Agency Issues                1.828912e+04
Graffiti                     2.327634e+04
Derelict Vehicle             2.535960e+04
Animal in a Park             1.212634e+06
Name: Request_Closing_Time, dtype: float64
```

Average Response Time

In [23]:
```
avg_time.plot(kind='bar', figsize=(12,5))
plt.title("Average Response Time")
plt.show()
```



Identify the significant variables by performing statistical analysis using p-values

In [25]:
```
from scipy.stats import f_oneway

groups = df.groupby('Complaint Type')['Request_Closing_Time'].apply(list)

anova_result = f_oneway(*groups)

print(anova_result)
```

```
F_onewayResult(statistic=np.float64(565.2615700417634), pvalue=np.float64(0.0))
```

Kruskal-WallisH

In [26]:
```
from scipy.stats import kruskal

kruskal_result = kruskal(*groups)

p = kruskal_result.pvalue

if p < 0.05:
    print("Reject H0: One or more distributions are not equal")
else:
    print("Fail to Reject H0: All distributions are equal")
```

```
Reject H0: One or more distributions are not equal
```

📌 Observations

- **Multiple Complaint Types:** Different categories of complaints are present across cities.
- **Common Issues:** Noise, illegal parking, and blocked driveways are most frequent.
- **Highest Complaints:** New York City has the maximum number of requests.
- **Response Time:** Varies depending on complaint type.
- **Resolution Time:** Some issues are resolved quickly, others take longer.
- **Statistical Result:** Significant difference in response time (p-value < 0.05).